Question Bank

Unit 1: Introduction and Physical Layer (15 Sessions)

Q1. Explain the different types of computer networks with suitable examples.

A **computer network** is a collection of interconnected computers and devices that communicate with each other to share data, resources, and services.

Computer networks can be classified mainly according to their **geographical coverage, size, and purpose**.

1. PAN (Personal Area Network)

A **Personal Area Network (PAN)** is a small network used to connect devices around a single person.

Range: Usually a few meters.

Examples:

- Smartphone connected to wireless earbuds through Bluetooth.
- Laptop connected to a wireless mouse.
- Smartphone sharing Internet with a laptop through hotspot.

Advantages:

- Easy to set up.
- Low cost.
- Requires little infrastructure.

Applications:

- Connecting personal devices.
- File transfer between nearby devices.
- Bluetooth peripherals.

2. LAN (Local Area Network)

A **Local Area Network (LAN)** connects computers and devices within a limited geographical area such as a room, home, office, school, or laboratory.

Examples:

- Computers connected in a college computer laboratory.
- Devices connected to a Wi-Fi router in a home.

Characteristics:

- High data transmission speed.
- Covers a relatively small area.
- Usually owned and managed by a single organization or person.

Applications:

- Sharing printers and files.
- Internet access.
- Communication between employees.
- Sharing centralized servers.

3. MAN (Metropolitan Area Network)

A **Metropolitan Area Network (MAN)** covers a larger area than a LAN, generally extending across a city or metropolitan region.

Example:
A network connecting different branches of a bank located throughout a city.

Characteristics:

- Larger coverage than LAN.
- Usually connects multiple LANs.
- Can be operated by organizations, telecom providers, or governments.

Applications:

- Connecting offices within a city.
- Cable television networks.
- City-wide Internet infrastructure.

4. WAN (Wide Area Network)

A **Wide Area Network (WAN)** covers a very large geographical area such as a country, continent, or the entire world.

The **Internet** is the largest example of a WAN.

Examples:

- A multinational company connecting offices in India, Germany, and the USA.
- Banking networks connecting branches across countries.

Characteristics:

- Covers very large geographical areas.
- Uses leased lines, fiber-optic links, satellites, microwave links, etc.
- Generally has higher latency than LAN.

Applications:

- International business communication.
- Internet services.
- Online banking.
- Cloud computing.

Comparison

Network	Full Form	Coverage	Example
PAN	Personal Area Network	Few meters	Bluetooth devices
LAN	Local Area Network	Building/ campus	College lab
MAN	Metropolitan Area Network	City	City-wide network
WAN	Wide Area Network	Country/world	Internet

Q2. Explain the different types of network topologies with suitable examples.

Network topology refers to the physical or logical arrangement of computers, network devices, and communication links in a network.

The major types are:

1. Bus Topology

In a **bus topology**, all devices are connected to a single main communication cable called the **backbone**.

```
Computer A ─┐
Computer B ─┼────────── Backbone ──────────┬─ Computer C
Computer D ─┘                              └─ Computer E
```

Advantages:

- Simple to install.
- Requires less cable.
- Relatively inexpensive.

Disadvantages:

- Failure of backbone can affect the entire network.
- Difficult to troubleshoot.
- Performance decreases as more devices are added.
- Limited cable length.

Example: Older Ethernet networks.

2. Star Topology

In a **star topology**, every device is connected to a central device such as a switch or hub.

```
             Computer A
                 |
                 |
Computer B ——— Switch ——— Computer C
                 |
                 |
             Computer D
```

Advantages:

- Easy to install and manage.
- Failure of one cable affects only one device.
- Easy to add or remove devices.
- Easy troubleshooting.

Disadvantages:

- Failure of central switch can affect the entire network.
- Requires more cable than bus topology.
- Central device increases cost.

Example: Modern Ethernet LANs.

3. Ring Topology

In a **ring topology**, each device is connected to two other devices, forming a closed loop.

```
Computer A ——— Computer B
     |              |
     |              |
Computer D ——— Computer C
```

Advantages:

- Data transmission can be organized systematically.
- No central device is required.

Disadvantages:

- Failure of one device/link may disrupt communication.
- Adding or removing devices can be difficult.
- Troubleshooting is comparatively difficult.

Example: Token Ring networks.

4. Mesh Topology

In a **mesh topology**, devices are connected to multiple or all other devices.

```
A ───── B
|\     /|
| \   / |
|  \ /  |
|  / \  |
| /   \ |
C ───── D
```

There are two forms:

- **Full Mesh:** Every device connects directly to every other device.
- **Partial Mesh:** Only selected devices have multiple connections.

Advantages:

- Highly reliable.
- Multiple paths are available.
- Failure of one link does not necessarily stop communication.

Disadvantages:

- Expensive.
- Requires large amounts of cabling.
- Complex installation and maintenance.

Example: Internet backbone networks.

5. Tree Topology

A **tree topology** combines characteristics of star and bus topologies. Devices are organized hierarchically.

```
            Core
          /      \
      Switch    Switch
     /   \        /   \
    A     B     C      D
```

Advantages:

- Scalable.
- Easy to manage hierarchically.
- Suitable for large networks.

Disadvantages:

- Failure of a higher-level device can affect connected segments.
- More complex than star topology.

Example: Large organizational networks.

6. Hybrid Topology

A **hybrid topology** combines two or more different topologies.

For example, an organization may use star topology within individual departments and connect departments using a mesh or tree structure.

Advantages:

- Flexible.
- Scalable.
- Can be designed according to organizational requirements.

Disadvantages:

- Expensive.
- Complex design and maintenance.

Q3. Give an overview of computer networks. Explain their basic components, advantages, and applications with suitable examples.

A **computer network** is a system in which two or more computers or devices are interconnected to communicate and exchange information.

Networks allow devices to **share data, hardware, software, Internet connectivity, and other resources**.

Basic Components of a Computer Network

1. Sender and Receiver

The **sender** generates and transmits data, while the **receiver** receives the data.

Example: A computer sending a file to another computer.

2. Transmission Medium

The transmission medium carries data from one device to another.

Examples:

- Ethernet cable
- Fiber-optic cable
- Radio waves
- Microwave

3. Network Interface Card

A **Network Interface Card (NIC)** allows a computer or device to connect to a network.

Modern computers commonly have Ethernet and/or Wi-Fi network interfaces.

4. Switch

A **switch** connects multiple devices in a LAN and forwards data to the appropriate destination.

5. Router

A **router** connects different networks and determines appropriate paths for forwarding packets.

For example, a home router connects the home LAN to the Internet.

6. Protocols

Protocols are rules that define how devices communicate.

Examples:

- TCP
- IP
- HTTP
- DNS
- Ethernet

Advantages of Computer Networks

1. **Resource sharing:** Printers, storage, and Internet connections can be shared.
2. **File sharing:** Users can exchange files easily.
3. **Communication:** Email, messaging, and video conferencing are possible.
4. **Centralized management:** Data and applications can be managed centrally.
5. **Cost reduction:** Shared resources reduce infrastructure costs.
6. **Remote access:** Users can access systems from different locations.
7. **Scalability:** Additional devices can be connected when required.

Applications

Computer networks are used in:

- Banking
- Education
- Healthcare
- E-commerce
- Cloud computing
- Social media
- Online gaming
- Government services
- Industrial automation
- Remote work

Example: In a university, computers in laboratories can connect to a central server, share printers, access the Internet, and access common educational resources.

Q4. Explain the OSI (Open Systems Interconnection) reference model. Describe its seven layers and their functions with suitable examples.

The **OSI (Open Systems Interconnection) model** is a conceptual reference model developed by the **International Organization for Standardization (ISO)**.

It divides network communication into **seven layers**, with each layer performing specific functions.

```
+----------------------------+
| 7. Application             |
+----------------------------+
| 6. Presentation            |
+----------------------------+
| 5. Session                 |
+----------------------------+
| 4. Transport               |
+----------------------------+
| 3. Network                 |
+----------------------------+
| 2. Data Link               |
+----------------------------+
| 1. Physical                |
+----------------------------+
```

1. Physical Layer

The Physical Layer is responsible for transmitting **raw bits** over the communication medium.

Functions:

- Bit transmission.
- Electrical/optical/radio signaling.
- Defines cables, connectors, frequencies, and transmission characteristics.

Examples:

- Ethernet cables.
- Fiber-optic cables.
- Radio signals.
- Network connectors.

2. Data Link Layer

The Data Link Layer provides **node-to-node delivery** and organizes bits into frames.

Functions:

- Framing.
- MAC addressing.
- Error detection.
- Flow control.
- Medium access control.

Examples:

- Ethernet.
- Wi-Fi MAC.
- Switches.

3. Network Layer

The Network Layer is responsible for **logical addressing and routing**.

Functions:

- IP addressing.
- Routing.
- Packet forwarding.
- Path determination.

Example:
IP protocol.

Routers primarily operate at this layer.

4. Transport Layer

The Transport Layer provides **end-to-end communication** between applications.

Functions:

- Segmentation.
- Flow control.
- Error recovery.
- Reliable delivery.
- Port addressing.

Examples:

- TCP
- UDP

5. Session Layer

The Session Layer establishes, manages, and terminates communication sessions between applications.

Functions:

- Session establishment.
- Session management.
- Synchronization.
- Session termination.

Example: Maintaining a logical communication session between applications.

6. Presentation Layer

The Presentation Layer deals with the **format and representation of data**.

Functions:

- Data translation.
- Encryption/decryption.
- Compression/decompression.
- Character encoding.

Examples:

- JPEG
- ASCII
- UTF-8
- Encryption formats

7. Application Layer

The Application Layer provides network services directly to user applications.

Functions:

- Web communication.
- Email.
- File transfer.
- Name resolution.

Examples:

- HTTP/HTTPS
- FTP
- SMTP
- DNS

Data Flow

At the sender:

Application → Presentation → Session → Transport → Network → Data Link → Physical

At the receiver, the reverse process occurs.

Q5. Explain the TCP/IP model. Describe its four layers, their functions, and give suitable examples.

The **TCP/IP model** is the networking model used as the foundation of the Internet.

It consists of **four layers**:

```
+------------------------------+
| Application                  |
+------------------------------+
| Transport                    |
+------------------------------+
| Internet                     |
+------------------------------+
| Network Access / Link        |
+------------------------------+
```

1. Application Layer

The Application Layer provides network services to user applications.

It combines the functions of the OSI **Application, Presentation, and Session layers**.

Functions:

- Web communication.
- Email.
- File transfer.
- Domain-name resolution.
- Remote access.

Protocols:

- HTTP/HTTPS
- DNS
- FTP
- SMTP
- SSH

2. Transport Layer

This layer provides communication between applications running on different hosts.

Functions:

- Segmentation.
- Port addressing.
- Flow control.
- Error control.
- Reliable or unreliable delivery.

Protocols:

- TCP
- UDP

TCP provides reliable, connection-oriented communication.

UDP provides faster, connectionless communication without TCP's reliability mechanisms.

3. Internet Layer

The Internet Layer handles logical addressing and routing of packets.

Functions:

- IP addressing.
- Routing.
- Packet forwarding.
- Internetworking.

Protocols:

- IPv4
- IPv6
- ICMP

Routers primarily operate at this layer.

4. Network Access / Link Layer

This layer handles communication over the physical network.

Functions:

- Framing.
- MAC addressing.
- Physical transmission.
- Access to the transmission medium.

Examples:

- Ethernet
- Wi-Fi

- Fiber networks

Example

When a user opens a website:

1. Application Layer uses HTTPS.
2. Transport Layer uses TCP.
3. Internet Layer adds IP addressing.
4. Link Layer places the packet into an Ethernet/Wi-Fi frame.
5. Data is transmitted over the physical medium.

Q6. Compare the OSI and TCP/IP reference models.

Feature	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Number of layers	7	4
Developed by	ISO	DARPA/DoD research ecosystem
Nature	Reference/conceptual model	Practical protocol suite and model
Application layers	Application, Presentation, Session	Application
Transport	Transport	Transport
Network	Network	Internet
Data Link + Physical	Separate	Combined into Link/Network Access
Protocol dependency	Protocol-independent	Built around TCP/IP protocols
Practical use	Mainly educational/ reference	Widely used in real networks
Development approach	Model first, protocols later	Protocols and architecture evolved together

Layer Mapping

```
OSI                         TCP/IP

Application    ┐
Presentation   ├────────── Application
Session        ┘

Transport ─────────────── Transport
```

```
Network  ─────────────────────  Internet

Data Link ──┐
Physical  ──┴───────────────────  Network Access
```

Conclusion

The OSI model provides a more detailed conceptual separation of networking functions, while TCP/IP is the practical architecture underlying modern Internet communication.

Q7. Explain guided transmission media. Describe its types, advantages, disadvantages, and applications with suitable examples.

Guided transmission media are communication media in which signals travel through a **physical path or cable** from sender to receiver.

The major types are:

1. Twisted Pair Cable
2. Coaxial Cable
3. Fiber-Optic Cable

1. Twisted Pair Cable

It consists of two insulated copper wires twisted together to reduce electromagnetic interference.

Types:

- **UTP:** Unshielded Twisted Pair
- **STP:** Shielded Twisted Pair

Advantages:

- Low cost.
- Easy installation.
- Flexible.
- Widely available.

Disadvantages:

- Limited distance.
- Susceptible to interference, especially UTP.
- Lower bandwidth than fiber.

Applications:

- Telephone networks.
- Ethernet LANs.
- Office networks.

2. Coaxial Cable

A coaxial cable consists of:

- Central conductor.
- Insulating layer.
- Metallic shield.
- Outer protective covering.

```
+-------------------------+
| Protective outer layer  |
| +---------------------+ |
| | Metallic shield     | |
| | +-----------------+ | |
| | | Insulator       | | |
| | | +-------------+ | | |
| | | | Conductor   | | | |
| | | +-------------+ | | |
| | +-----------------+ | |
| +---------------------+ |
+-------------------------+
```

Advantages:

- Better shielding than twisted pair.
- Higher bandwidth than basic twisted pair.
- More resistant to interference.

Disadvantages:

- More expensive than twisted pair.
- Less flexible.
- Installation is more difficult.

Applications:

- Cable television.
- Broadband networks.
- CCTV systems.

3. Fiber-Optic Cable

Fiber-optic cable transmits information using **light signals** through thin strands of glass or plastic.

Types:

- Single-mode fiber.
- Multimode fiber.

Advantages:

- Very high bandwidth.
- Long-distance transmission.
- Low signal loss.
- Immune to electromagnetic interference.
- More secure against electromagnetic interception.

Disadvantages:

- Higher installation cost.
- Fragile compared with copper.
- Requires specialized equipment and expertise.

Applications:

- Internet backbone.
- Long-distance telecommunications.
- Data centers.
- Submarine communication cables.

Q8. Explain Unguided transmission media. Describe its types, advantages, disadvantages, and applications with suitable examples.

Unguided transmission media, also called **wireless transmission media**, transmit signals through air or space without requiring a physical cable.

Major types include:

1. Radio waves
2. Microwaves
3. Infrared
4. Satellite communication

1. Radio Waves

Radio waves can travel through air and can cover relatively large areas.

Advantages:

- No physical cable required.
- Can cover large areas.
- Suitable for mobile communication.

Disadvantages:

- Interference.
- Security concerns.
- Signal quality may vary.

Applications:

- Radio broadcasting.
- Wi-Fi.
- Mobile communication.
- IoT devices.

2. Microwaves

Microwaves use high-frequency electromagnetic waves for communication. Terrestrial microwave communication generally requires **line of sight**.

Advantages:

- High bandwidth.
- Suitable for long-distance communication.
- No physical cable required.

Disadvantages:

- Requires line of sight.
- Buildings and terrain can obstruct signals.
- Weather can affect some microwave links.

Applications:

- Cellular networks.
- Point-to-point communication.
- Television transmission.

3. Infrared

Infrared communication uses infrared light for short-range communication.

Advantages:

- Low interference between rooms.
- Relatively secure over short distances.
- Inexpensive.

Disadvantages:

- Short range.
- Cannot normally pass through walls.
- Requires suitable alignment in many applications.

Applications:

- Remote controls.
- Short-range device communication.

- Sensors.

4. Satellite Communication

Satellite communication uses satellites as communication relays between ground stations.

```
Ground Station A
       ↑
       |
   [Satellite]
       |
       ↓
Ground Station B
```

Advantages:

- Very large coverage area.
- Useful for remote regions.
- Suitable for broadcasting.

Disadvantages:

- High cost.
- Higher propagation delay.
- Weather and atmospheric conditions can affect certain frequency bands.

Applications:

- Satellite TV.
- GPS/GNSS services.
- Weather monitoring.
- Remote communication.
- Global telecommunications.

Q9. Differentiate between guided and unguided transmission media.

Feature	Guided Media	Unguided Media
Meaning	Uses physical path	Uses air/space
Transmission	Through cables	Through electromagnetic waves
Examples	Twisted pair, coaxial, fiber	Radio, microwave, infrared
Mobility	Limited	High
Installation	Requires physical cabling	Usually no physical cable

Interference	Generally lower, especially fiber	More susceptible
Security	Generally easier to physically control	Wireless signals can be intercepted
Bandwidth	Fiber provides extremely high bandwidth	Depends on frequency and technology
Coverage	Cable-dependent	Can cover large areas
Cost	Cabling can be expensive	Deployment may be cheaper for some applications
Example	Ethernet cable	Wi-Fi

Conclusion

Guided media are suitable where a stable physical connection and high reliability are required, while unguided media are useful when **mobility, flexibility, and wide-area wireless connectivity** are important.

Unit 2: Data Link Layer (15 Sessions)

Q1. Define character stuffing in computer networks. Explain how it is used in characteroriented framing with a suitable example.

Character stuffing is a technique used in **character-oriented framing** to distinguish control characters from actual data.

In character-oriented protocols, special characters are used to indicate the **beginning and end of a frame**.

Suppose:

- `FLAG` indicates the beginning/end of a frame.

- `ESC` is an escape character.

If the actual data contains `FLAG` or `ESC`, the sender inserts an `ESC` before it. This process is called **character stuffing**.

Example

Suppose the frame delimiter is:

```
FLAG
```

Original data:

```
A B FLAG C
```

The receiver could incorrectly interpret `FLAG` as the end of the frame.

Therefore, the sender stuffs an escape character:

```
FLAG A B ESC FLAG C FLAG
```

At the receiver, when `ESC FLAG` is encountered, it understands that `FLAG` is **data**, not a frame delimiter.

The receiver removes the extra `ESC` character and reconstructs:

```
A B FLAG C
```

Process

Sender:

```
Data → Search for special characters → Insert ESC → Send
frame
```

Receiver:

```
Receive frame → Detect ESC → Remove ESC → Recover
original data
```

Advantages

- Simple technique.
- Prevents confusion between control characters and data.
- Useful for character-oriented protocols.

Disadvantage

The size of the transmitted frame increases when data contains many special characters.

Q2. Define bit stuffing in computer networks. Explain how it is used in bitsoriented framing with a suitable example.

Bit stuffing is a framing technique used in **bit-oriented protocols** to ensure that a special bit pattern used as a frame delimiter does not accidentally appear in the data.

A common flag pattern is:

```
01111110
```

The sender follows this rule:

Whenever five consecutive `1`s occur in the data, the sender inserts a `0`.

The receiver removes the inserted `0` after detecting five consecutive `1`s.

Example

Suppose original data is:

```
011111101
```

The data contains five consecutive `1`s:

```
0 11111 0 1
```

The sender inserts `0` after the five `1`s:

```
0111110101
```

The frame can then be transmitted as:

```
FLAG + Stuffed Data + FLAG
```

```
01111110  0111110101  01111110
```

At the receiver, after five consecutive `1`s, the inserted `0` is removed.

```
0111110101
      ↑
  stuffed 0
```

The original data is recovered:

```
011111101
```

Advantages

- Prevents accidental occurrence of the flag pattern in data.
- Provides reliable frame boundary detection.
- Used in bit-oriented protocols such as HDLC.

Difference from Character Stuffing

Character stuffing works with **characters/bytes**, whereas bit stuffing works at the **individual bit level**.

Q3. Describe Vertical Redundancy Check (VRC), Longitudinal Redundancy Check (LRC with examples.

Vertical Redundancy Check (VRC)

VRC, commonly called **parity checking**, adds one extra parity bit to each transmitted data unit.

The parity bit makes the total number of `1`s either:

- **Even** for even parity.
- **Odd** for odd parity.

Example: Even Parity

Suppose data is:

```
1011001
```

Number of 1s = 4, which is already even.

Therefore:

```
Data:        1011001
Parity bit: 0
Transmitted:10110010
```

If the receiver receives:

```
10110011
```

there are now 5 ones, which is odd.

Therefore, an error is detected.

Advantages

- Simple.
- Low overhead.
- Easy to implement.

Disadvantage

VRC may fail to detect errors involving an **even number of bit changes**.

Longitudinal Redundancy Check (LRC)

LRC calculates parity across multiple data units, usually arranged in rows and columns.

An additional parity row is generated so that each column satisfies the selected parity condition.

For example:

```
      1  0  1  1
      1  0  1  0
      1  1  1  1
      1  0  1  0
      --------
LRC   0  1  0  0
```

The LRC bits are calculated column-wise.

For even parity:

Column 1

```
1  1  1  1
```

Number of 1s = 4

LRC bit = 0

Column 2

```
0  0  1  0
```

Number of 1s = 1

LRC bit = 1

Column 3

```
1  1  1  1
```

Number of 1s = 4

LRC bit = 0

Column 4

```
1  0  1  0
```

Number of 1s = 2

LRC bit = 0

Therefore:

```
LRC = 0100
```

LRC can detect many errors that simple VRC cannot, particularly when errors affect different rows and columns.

Q4. The sender transmits the following byte code to the receiver. Using the Vertical Redundancy Check (VRC) method, demonstrate that an error has occurred in the received byte code.

Sr No	Sender side	Receiver side
1	1 0 1 1 1 0 1 0	1 0 1 1 1 0 1 1
2	1 1 1 1 1 0 1 0	1 1 0 1 1 0 1 0
3	1 1 0 0 1 0 1 0	1 1 1 0 1 0 1 1
4	0 1 0 0 1 0 1 0	0 1 0 0 1 0 1 0

Q5. The sender transmits the following byte code to the receiver. Using the Longitudinal Redundancy Check (LRC) method, demonstrate that an error has occurred in the received byte code.

Sr No	Sender side	Receiver side
1	1 0 1 1 1 0 1 0 1 1 1 1 1 0 1 0	1 0 1 1 1 0 1 0 1 1 0 1 1 0 1 0
2	1 1 0 0 1 0 1 0 0 1 0 1 0 1 0 1	1 1 0 0 1 0 1 0 0 1 0 1 0 1 1 1
3	1 1 0 1 1 0 1 0 0 1 0 1 1 1 0 1	1 1 0 1 1 0 1 0 0 1 0 1 1 1 0 1